

# Ali R. Vahdati, Ph.D.

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🏠 1989    📍 Zurich, Switzerland



## Work Experience

- 2020 – 2023    📌 **Senior lecturer.** University of Zurich, Switzerland  
\* Conducted extensive research on ancient human dispersals and extinctions, demonstrating strong analytical and problem-solving skills.
- 2022 – 2023    📌 **Modeling Consultant.** University of Madeira, Portugal.  
\* Constructed a model for understanding metacommunity dynamics of complex life-cycles in exploited seascapes.
- May and June 2022    📌 **Consultant.** IBS center for climate physics, University of Pusan, South Korea.  
\* Developed a comprehensive agent-based model, demonstrating my strong skills in computational modeling and programming.
- 2017 – 2020    📌 **Postdoctoral researcher.** University of Zurich, Switzerland.  
\* Led multiple research projects, demonstrating strong leadership skills and project management capabilities.
- Oct. 2021    📌 **Modeling Consultant.** Centro de investigación Científica de Yucatán, Mérida, Mexico.  
\* Developed statistical models to understand interaction strengths in food webs.

## Skills

- Predictive Modeling    📌 Proficient in creating and interpreting complex biological models. Applied these skills to understand and predict biological phenomena, contributing to advancements in fields such as human evolution, population genetics, and ecology.
- Data-Science and Analytics    📌 Experienced in applying machine learning and deep learning to understand large datasets and make predictions.
- Statistics    📌 Utilized advanced statistical methods to interpret and analyze data from various biological research projects, facilitating informed decision-making.
- Agent-based Modeling    📌 Developed agent-based models to simulate and analyze complex biological systems, contributing to a deeper understanding of phenomena such as human genetic diversity and ancient human dispersals.
- Experience with Large Datasets    📌 Experienced in handling and analyzing large genomic datasets from high-throughput technologies, facilitating the extraction of meaningful insights and advancing research in evolutionary biology.
- Population Genetics    📌 Applied knowledge of population genetics in various research projects, contributing to the understanding of genetic diversity and evolution.

## Skills (continued)

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|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming         | Proficient in multiple programming languages and tools including Julia, Python, R, LaTeX, MySQL, SQLite, Git, Continuous integration, Unit testing, and Docker. Experience in creating reproducible research and statistical programming. Familiar with Linux command line and high-performance computing environments, enabling the development of efficient algorithms and models for biological research. |
| Soft Skills         | Strong analytical thinking and problem-solving skills, adaptability to different business challenges and data sources, as well as curiosity and openness to new solutions.                                                                                                                                                                                                                                   |
| Transferable Skills | Proven ability in teaching, presentation, and writing, with experience in effectively communicating complex data and research findings to diverse audiences including biologists, clinicians, and data scientists. Proficient in time management and project management, enhancing the effectiveness of research processes and facilitating knowledge transfer in academic settings.                         |

## Education

- 2012 – 2017 **Ph.D., University of Zurich** Evolutionary Biology.  
Thesis title: *Thesis: Genotype networks: convergent evolution, population size and adaptation.*
- 2010 – 2011 **M.Sc, University of Manchester** n Evolutionary Biology and Genomics.  
Thesis title: *Analysis of replication timing on genetic variation in the human genome..*

## Research Publications

### Journal Articles

- 1 G. Datseris, **A. R. Vahdati**, and T. C. DuBois, “Agents.jl: A performant and feature-full agent-based modeling software of minimal code complexity,” *Simulation*, p. 003 754 972 110 688, Jan. 2022, \_eprint: 2101.10072, ISSN: 0037-5497.  DOI: 10.1177/00375497211068820.
- 2 **A. R. Vahdati** and C. J. Melián, “EvoDynamics.jl: A framework for modeling eco-evolutionary dynamics,” en, *Journal of Open Source Software*, vol. 7, no. 79, p. 4775, Nov. 2022, ISSN: 2475-9066.  DOI: 10.21105/joss.04775. (visited on 11/18/2022).
- 3 **A. R. Vahdati**, J. D. Weissmann, A. Timmermann, M. Ponce de León, and C. P. Zollikofer, “Exploring Late Pleistocene hominin dispersals, coexistence and extinction with agent-based multi-factor models,” *Quaternary Science Reviews*, vol. 279, p. 107 391, Mar. 2022, ISSN: 02773791.  DOI: 10.1016/j.quascirev.2022.107391.
- 4 **A. R. Vahdati**, “Agents.jl: Agent-based modeling framework in Julia,” *Journal of Open Source Software*, vol. 4, no. 42, p. 1611, Oct. 2019, ISSN: 2475-9066.  DOI: 10.21105/joss.01611.
- 5 **A. R. Vahdati**, J. D. Weissmann, A. Timmermann, M. S. Ponce de León, and C. P. Zollikofer, “Drivers of Late Pleistocene human survival and dispersal: An agent-based modeling and machine learning approach,” *Quaternary Science Reviews*, vol. 221, p. 105 867, Oct. 2019, ISSN: 02773791.  DOI: 10.1016/j.quascirev.2019.105867.
- 6 **A. R. Vahdati** and A. Wagner, “Population size affects adaptation in complex ways: Simulations on empirical adaptive landscapes,” *Evolutionary Biology*, vol. 45, no. 2, pp. 156–169, Jun. 2018, ISBN: 0123456789 Publisher: Springer US, ISSN: 0071-3260.  DOI: 10.1007/s11692-017-9440-9.

- 7 **A. R. Vahdati**, K. Sprouffske, and A. Wagner, "Effect of population size and mutation rate on the evolution of RNA sequences on an adaptive landscape determined by RNA folding," *International Journal of Biological Sciences*, vol. 13, no. 9, pp. 1138–1151, 2017, ISSN: 1449-2288.  DOI: 10.7150/ijbs.19436.
- 8 **A. R. Vahdati** and A. Wagner, "Parallel or convergent evolution in human population genomic data revealed by genotype networks," *BMC Evolutionary Biology*, vol. 16, no. 1, p. 154, Dec. 2016, ISSN: 1471-2148.  DOI: 10.1186/s12862-016-0722-0.
- 9 G. M. Dall'Olio, **A. R. Vahdati**, J. Bertranpetit, A. Wagner, and H. Laayouni, "VCF2Networks: Applying genotype networks to single-nucleotide variants data," *Bioinformatics*, vol. 31, no. 3, pp. 438–439, Feb. 2015, ISSN: 1460-2059.  DOI: 10.1093/bioinformatics/btu650.
- 10 S. Kärst, **A. R. Vahdati**, G. a. Brockmann, and R. Hager, "Genomic imprinting and genetic effects on muscle traits in mice.," *BMC genomics*, vol. 13, p. 408, Jan. 2012, ISSN: 1471-2164.  DOI: 10.1186/1471-2164-13-408.

## Selected Teaching Experience

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| 2019 – 2022 |  <b>Lecturer</b> , BIO397 - Applied Machine Learning - University of Zurich.<br>I have designed and delivered lectures and exercises on Julia programming language, linear algebra, multivariate calculus, machine learning algorithms, neural networks, and evaluation and improvement of machine learning models.<br>I have prepared students with no previous knowledge of machine learning to be able to design and perform complete machine learning workflows. |
| 2022        |  <b>Lecturer</b> , An introduction to agent-based modeling, University of Pusan, South Korea.<br>I gave a lecture and hands-on tutorial on using Julia language for building agent-based models.                                                                                                                                                                                                                                                                   |
| 2018        |  <b>Lecturer</b> , Reproducible research workshop, EAWAG, Switzerland.<br>I gave a lecture and hands-on tutorial on using Git version control software and collaborating using Github.                                                                                                                                                                                                                                                                             |
|             |  <b>Teaching assistant</b> , Morphometric analysis BIO208, University of Zurich.<br>Helped students understand and solve problems on analyzing fossils by adding landmarks to their 3D scans and comparing them.                                                                                                                                                                                                                                                   |
| 2013        |  <b>Teaching assistant</b> , Introduction to Bioinformatics (BIO390), University of Zurich.<br>As part of a team, tutored about 30 students in the Evolutionary Biology and Environmental Studies programs to help them solve their programming questions in Python.<br>Maintained a high retention rate in assigned classes.<br>Tutored students improved performance over a single grading period.                                                               |
|             |  <b>Teaching assistant</b> , Principles of Evolution (BIO351), University of Zurich.<br>Prepared teaching material for R programming. Prepared and delivered two lectures on "modeling population genetic simulation of natural selection and drift processes". Fielded questions from students on lecture topics.<br>Gave an R programming lecture and helped students solve non-programming conceptual problems.                                                 |

## References

Available on Request